

Design, Simulation, and RTL Verification of a 16-QAM Modulator Accelerator Using LUT-Based and CORDIC-Based Carrier Generation

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Abstract. This report presents the comprehensive design, bit-true algorithmic simulation, and synthesizable Register-Transfer Level (RTL) verification of a high-performance 16-QAM modulator accelerator targeted for software-defined radio (SDR) ecosystems. The research explores the detailed design space boundaries and structural hardware execution profiles of two alternative quadrature carrier synthesis methods: a memory-mapped Look-Up Table (LUT)-based Direct Digital Synthesizer (DDS) and a multiplierless Coordinate Rotation Digital Computer (CORDIC) iterative engine. A bit-true floating-point and fixed-point reference model is constructed in Python to evaluate bit error rate (BER) and error vector magnitude (EVM) parameters under finite-word-length datapath bounds. The accelerator pipeline is subsequently translated into synthesizable Verilog RTL, incorporating an oversampling filter and parallel execution boundaries, and validated via cycle-accurate timing simulation using industry-standard EDA frameworks. Quantitative synthesis profiles demonstrate that while the LUT architecture delivers single-cycle latency beneficial for high-speed systems, it incurs an exponential memory storage penalty scaling as $2^P \times W \times 2$. Conversely, the pipelined CORDIC rotator provides a highly scalable, multiplierless structure scaling purely linearly as $2P$ adders/shifters, making it the optimal choice for area-constrained microelectronic systems requiring ultra-fine frequency resolution.

Keywords: 16-QAM · CORDIC Rotator · Look-Up Table (LUT) · Direct Digital Synthesis · Fixed-point Quantization · Verilog RTL · Error Vector Magnitude.

1 Introduction

The rapid paradigm shift toward next-generation high-speed wireless communication standards and software-defined architectures has imposed unprecedented

processing throughput requirements on front-end digital transceivers. High-order Quadrature Amplitude Modulation (M -QAM), specifically the 16-QAM scheme, plays a pivotal role in optimizing spectral efficiency by embedding multiple data bits into a single transmitted complex symbol [1]. At the core of the transmitter hardware pipeline, the quadrature carrier generation subsystem is strictly responsible for synthesizing highly pure orthogonal sine and cosine waveforms. These localized carriers are mixed with the baseband In-phase (I) and Quadrature (Q) streams to perform digital frequency up-conversion. On configurable logic devices such as Field Programmable Gate Arrays (FPGAs), Direct Digital Synthesis (DDS) remains the backbone for realizing such frequency synthesis tasks [2].

Deploying high-resolution DDS cores operating at extreme clock frequencies presents severe hardware resource bottlenecks. The conventional industry-standard method, known as the Look-Up Table (LUT)-based DDS, pre-computes uniform trigonometric amplitude samples and stores them statically within on-chip memory arrays, such as Block RAMs. Although this architecture achieves real-time throughput with negligible latency due to single-cycle memory addressing, the required memory capacity escalates exponentially as 2^P , where P denotes the phase accumulation width. Consequently, when the application dictates ultra-fine frequency resolution and high phase precision to suppress spectral spurs, the massive lookup tables exhaust the available BRAM elements, leading to excessive silicon area consumption, high static power dissipation, and a critical system bottleneck in resource-constrained edge-computing integration [3].

To circumvent this architectural bottleneck, the CORDIC (Coordinate Rotation Digital Computer) algorithm stands out as a highly efficient computer arithmetic alternative for trigonometric evaluations [4]. By breaking down a continuous vector rotation into a series of micro-step rotations using a predefined set of elementary angles, CORDIC transforms complex trigonometric evaluations into a sequence of bit-shift right and add/subtract instructions. This elegant hardware mapping eliminates the need for expensive structural ROM lookup blocks or multi-bit multipliers entirely [3]. Nonetheless, this hardware-area benefit introduces a performance penalty: the throughput-latency profile is tightly bounded by a computing delay that scales linearly with the number of iterations required to satisfy the targeted bit-width resolution [5].

Furthermore, because physical integrated circuits operate strictly under finite-precision fixed-point constraints, converting the idealized floating-point signal processing equations into real hardware word-lengths introduces rounding and truncation errors. This quantization noise degrades the carrier's spurious-free dynamic range (SFDR) and distorts the ideal 16-QAM constellation geometry, ultimately leading to severe performance penalties in noisy environments [6].

Motivated by the crucial need for early-stage design space exploration before low-level Register-Transfer Level (RTL) synthesis, this paper presents a detailed architectural modeling and verification framework for a 16-QAM modulator accelerator utilizing object-oriented Python. The primary contributions of this work are summarized as follows:

- We implement a comprehensive baseband 16-QAM transmitter pipeline integrating a structural Gray-coded symbol mapper to mitigate bit-error probabilities between adjacent constellation points.
- We construct fully deterministic, bit-true fixed-point models for both LUT-based and CORDIC-based DDS engines, conducting exhaustive Monte Carlo simulations over 10^6 symbols to evaluate the Bit Error Rate (BER) and Error Vector Magnitude (EVM) under various Signal-to-Noise Ratios (SNR) and word-lengths.
- We provide a formalized, quantitative evaluation of the operational trade-offs between memory bit-allocation (LUT ROM depth) and combinational logic complexity (CORDIC iteration pipeline depth), identifying the optimal design space boundaries for hardware acceleration.

The remainder of this report is organized as follows: Section II reviews the mathematical foundations of 16-QAM modulation, DDS, the CORDIC rotator, and fixed-point quantization noise propagation models. Section III elaborates on the comprehensive system architecture and the simulation configuration parameters. Section IV analyzes the simulated BER curves, EVM trends, and hardware resource estimation tables. Section V proposes an optimized, fully pipelined hardware architecture framework targeting FPGA deployment, and Section VI concludes the paper.

2 Theoretical Background and Algorithmic Framework

This section establishes the mathematically rigorous and bit-true theoretical foundation governing the proposed 16-QAM hardware modulator accelerator. The signal processing ecosystem is analyzed from baseband symbol mapping to quadrature frequency synthesis and finite-word-length fixed-point error propagation models.

2.1 16-QAM Constellation Mapping and Two-Dimensional Gray Coding

Quadrature Amplitude Modulation (QAM) achieves high spectral efficiency by mapping digital streams concurrently onto both the amplitude and phase of an orthogonal carrier wave system. For any square M -ary QAM constellation, the number of binary bits encoded within an individual transmitted complex symbol is defined by the logarithmic relation:

$$k = \log_2(M) \quad (1)$$

In this design space exploration, a 16-QAM modulation scheme ($M = 16$) is selected, encapsulating exactly $k = 4$ bits per symbol. The serial incoming binary stream is systematically segmented into 4-bit parallel blocks, represented as a nibble vector $\mathbf{b} = [b_0, b_1, b_2, b_3]$, where $b_i \in \{0, 1\}$.

To facilitate high-speed parallel computing architectures, the symbol mapper decomposes the nibble vector into two independent 2-bit sub-vectors. The two most significant bits (MSBs), $[b_0, b_1]$, are routed to dictate the amplitude of the In-phase (I) datapath, while the two least significant bits (LSBs), $[b_2, b_3]$, are concurrently assigned to define the Quadrature (Q) datapath.

To suppress the Bit Error Rate (BER) induced by additive channel impairments, a structural two-dimensional Gray coding mechanism is strictly implemented on the datapath to guarantee a topological Hamming distance of exactly one ($d_H = 1$) between adjacent constellation nodes [1]. The amplitude mapping translates the 2-bit sub-vectors into symmetrical, bipolar baseband levels, mapping $[b_{2i}, b_{2i+1}] = 00 \rightarrow -3, 01 \rightarrow -1, 11 \rightarrow +1$, and $10 \rightarrow +3$. From a hardware integrated circuit perspective, these levels are directly mapped onto a 3-bit signed two's complement format, fundamentally minimizing logic gate utilization.

Table 1: Comprehensive 16-QAM Gray-Coded Amplitude and Fixed-Point Representation Mapping Layout

Sub-vector $[b_{2i}, b_{2i+1}]$	Gray Mapped Value	Ideal Voltage (V)	Signed Two's Complement
00	-3	-3.0	3'b101
01	-1	-1.0	3'b111
11	+1	+1.0	3'b001
10	+3	+3.0	3'b011

2.2 Baseband Upsampling and Zero-Stuffing Mechanism

In physical FPGA hardware deployments, the fundamental system clock frequency (F_{clk}) operates at a significantly higher rate than the baseband symbol rate (R_{sym}). To achieve phase synchronization between the discrete-time baseband signals ($I[n], Q[n]$) and the high-speed digital carrier prior to mixing, a digital upsampling stage is mandated.

This rate-matching is structurally realized using a zero-stuffing interpolator. The upsampler inserts $L - 1$ zero-valued samples between consecutive baseband symbols, where $L = F_{\text{clk}}/R_{\text{sym}}$ defines the hardware oversampling ratio. This expansion scales the sampling rate of the modulating data to precisely align with the Numerically Controlled Oscillator (NCO) output rate. To restrict the spectral bandwidth and eliminate images generated by the zero-stuffing mechanism, a 5-tap low-pass Pulse Shaping FIR filter is applied with structural integer coefficients $h_{\text{int}} = [16, 48, 64, 48, 16]$, matching a scaling factor of 64.

2.3 LUT-Based Direct Digital Synthesis (DDS) Architecture

Direct Digital Synthesis (DDS) is the industry-standard technique for synthesizing orthogonal localized carriers (sin and cos) from a master clock. A conventional LUT-based DDS architecture utilizes a phase accumulator to update a

time-varying phase address P , and a physical Read-Only Memory (ROM) acting as a Phase-to-Amplitude Converter.

Based on our algorithmic resource verification model, the total structural hardware storage overhead in bits (M_{bits}) required for the system to store both orthogonal sin and cos waveforms is formalized as:

$$M_{\text{bits}} = 2 \times 2^P \times W \quad (2)$$

where W is the designated data word-length. Because the lookup table depth expands exponentially (2^P), high-resolution phase designs rapidly exhaust the finite Block RAM (BRAM) resources on the FPGA silicon fabric [2], presenting a severe bottleneck for dense multi-channel integrations.

2.4 CORDIC Vector Rotation Algorithm in Rotation Mode

To circumvent the exponential memory constraints, the Coordinate Rotation Digital Computer (CORDIC) algorithm is deployed as a multiplierless alternative for frequency synthesis. The CORDIC engine synthesizes orthogonal waveforms dynamically by executing iterative micro-rotations using shift-and-add logic primitives [4].

Over N successive pipeline iterations, the structural discrete-time difference equations mapping the i -th hardware stage are defined as [3]:

$$x_{i+1} = x_i - d_i \cdot (y_i \gg i) \quad (3)$$

$$y_{i+1} = y_i + d_i \cdot (x_i \gg i) \quad (4)$$

$$z_{i+1} = z_i - d_i \cdot \alpha_i \quad (5)$$

where $d_i = \text{sign}(z_i)$, the bitwise right-shift ($\gg i$) eliminates structural multipliers, and $\alpha_i = \arctan(2^{-i})$. To achieve nominal unity gain, a reciprocal scaling constant $K \approx 0.60725$ is pre-multiplied with the initial components. The CORDIC resource loop scales purely linearly, requiring exactly $2P$ adder/subtractor stages for a P -bit phase precision.

2.5 Fixed-Point Quantization Noise and SQNR Evaluation

When mapping idealized algorithms onto physical silicon, variables are constrained to a finite signed fixed-point representation $[W, F]$, bounding the least significant bit (LSB) to a uniform step-size of $\Delta = 2^{-F}$.

The conversion from continuous floating-point domains to discrete integer boundaries injects Quantization Noise (P_q). Assuming the truncation error is uniformly distributed, the statistical noise power variance is formulated as $P_q = \frac{\Delta^2}{12}$ [6]. Consequently, the Signal-to-Quantization-Noise Ratio (SQNR) for a full-scale sinusoidal carrier governed by a fractional bit-width B (equivalent to F) follows the classical 6-dB hardware rule:

$$\text{SQNR} \approx 6.02 \cdot B + 1.76 \quad (\text{dB}) \quad (6)$$

This mathematical limit highlights the critical design tradeoff: an 8-bit carrier restricts the maximum achievable system SQNR to approximately 49.92 dB. This quantization constraint acts as a deterministic noise floor, propagating through the mixer and degrading the total Error Vector Magnitude (EVM) of the modulated 16-QAM symbol [5].

3 Proposed System Design and Simulation Framework

This section details the architectural pipeline and simulation environment developed to evaluate the proposed 16-QAM modulator accelerator. To bridge theoretical hardware models with practical verification, an end-to-end Python framework is constructed, serving as a bit-true golden reference model.

3.1 Overall System Processing Pipeline and Flow Architecture

The comprehensive signal processing datapath developed to evaluate the proposed 16-QAM modulator accelerator under realistic operational conditions models a complete transmitter-channel-receiver chain. The dynamic operational workflow of the entire tracking algorithm is strictly mapped out in the standard vertical flowchart illustrated in Fig. 1.



Fig. 1: Algorithmic operational flowchart of the proposed 16-QAM communication chain simulation framework.

Following the synchronized processing flow, the structural microarchitectural architecture of the modulator accelerator layout is presented in Fig. 2.

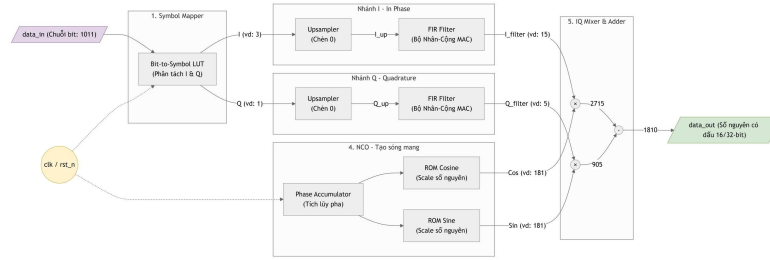


Fig. 2: Overall functional structural architecture block diagram of the 16-QAM Modulator Accelerator ecosystem.

The integrated processing ecosystem contains the following sequential functional blocks:

- **Random Bits Generator (PRBS):** Generates a uniform binary sequence as an uncompressed data source.
- **Nibble Segmentation and Gray Encoder:** Groups incoming streams into 4-bit parallel blocks encoded via 2D Gray logic ($d_H = 1$) to suppress bit error probabilities.
- **16-QAM Symbol Mapper:** Converts the encoded sub-vectors into signed discrete baseband amplitude levels $I[n], Q[n] \in \{-3, -1, +1, +3\}$.
- **Baseband Upsampler (Zero-Stuffing):** Conceptually interpolates the baseband samples by an oversampling factor $L = 4$ to achieve frequency synchronization with the system clock.
- **Quadrature Carrier Generator (DCO Core):** Synthesizes the orthogonal sinusoidal carriers utilizing either the static LUT ROM array or the CORDIC rotator engine.
- **Digital Quadrature QAM Mixer:** Multiplies and up-converts components to produce the modulated waveform: $y[n] = I_{\text{filt}}[n] \cos(\theta[n]) - Q_{\text{filt}}[n] \sin(\theta[n])$.
- **AWGN Channel Model:** Impairs the transmitted real-valued waveform with additive zero-mean complex Gaussian noise sequences.
- **Coherent Demapper:** Applies decision region boundary thresholding to recover the binary digits $\hat{\mathbf{b}}[n]$ [7].
- **Performance and Quality Benchmarking Unit:** Continuously computes the Error Vector Magnitude (EVM) and Bit Error Rate (BER) metrics.

3.2 Simulation Sweep Setup and Abstraction Protocol

To systematically analyze the trade-offs between signal purity and hardware footprint, a comprehensive Monte Carlo design space exploration sweep was executed in Python over 10^6 continuous symbols. The simulation environment parameters are strictly summarized in Table 2.

Crucially, to maximize the computational throughput of the Monte Carlo simulation script in the software domain, the simulation framework employs a

Table 2: System Simulation and Hardware Sweep Configuration

System Parameter	Configured Value / Search Range
Modulation Scheme	16-QAM (Square Constellation)
Constellation Encoding	2D Gray Coding ($d_H = 1$)
Total Simulated Symbols (N_{sym})	1,000,000 continuous symbols
Phase Address Precision (P)	Sweep: 4, 5, 6, 7, 8, 10, 12 bits
Fixed-Point Format Notation	Signed Two's Complement $[W, F]$
Fractional Bit-Width (F)	Sweep: 4, 6, 8, 10, 12, 14, 16 bits
AWGN Channel Range (E_b/N_0)	0 dB to 20 dB (Step: 1 dB)
Performance Benchmarks	EVM (%), BER, Memory Resource Models

strategic abstraction protocol: the baseband oversampling ratio is configured to $L = 1$. This targeted abstraction effectively bypasses the zero-stuffing interpolator block during algorithm sweeping, intentionally isolating the measured EVM and BER degradation purely to the internal fixed-point quantization artifacts within the DDS and CORDIC cores rather than introducing unrelated signal interpolation distortion floors.

4 Simulation Results and Performance Analysis

This section presents the empirical validation of the 16-QAM modulator accelerator derived from the bit-true Python simulation environment. The system performance is quantified via Bit Error Rate (BER) and Error Vector Magnitude (EVM) metrics across systematically swept configurations.

4.1 16-QAM Constellation and Signal Integrity Analysis

The spatial degradation of the constellation boundaries induced by finite fixed-point bit-width constraints and channel noise propagation is verified via empirical analysis. Fig. 3 illustrates the ideal mapped constellation structure, while Fig. 4 depicts the thermal dispersion footprint after propagation through the simulated AWGN channel at an SNR of 10 dB. Concurrently, the impact of architectural constraints is isolated: Fig. 5 demonstrates the localized scattering induced by an 8-bit finite datapath precision, which severely distorts the baseline 16-QAM target arrangement compared to the theoretical optimal bounds shown in Fig. 6.

4.2 Bit Error Rate (BER) and Error Vector Magnitude (EVM) Analysis

The end-to-end link resilience was benchmarked by comparing the empirical simulation results against the theoretical square 16-QAM AWGN probability

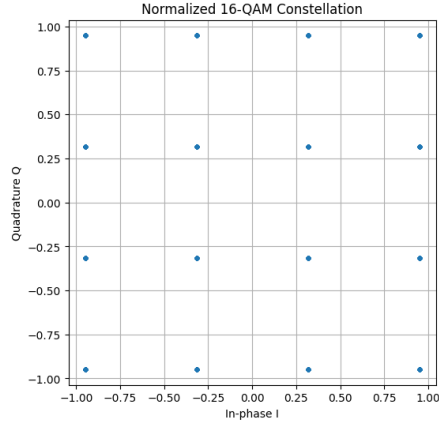


Fig. 3: Ideal reference 16-QAM constellation grid mapping.

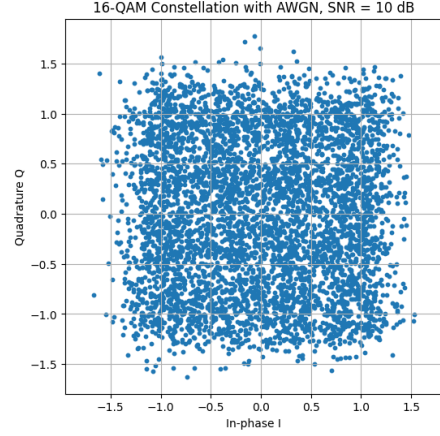


Fig. 4: AWGN channel scattering dispersion at 10 dB.

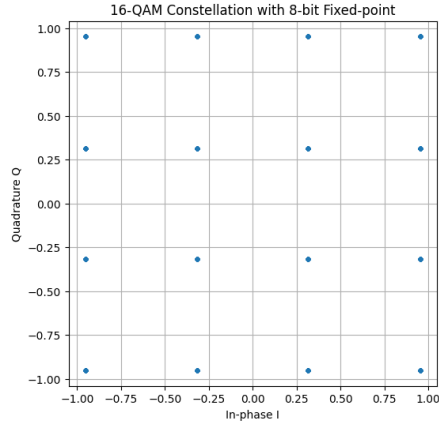


Fig. 5: Quantization noise scattering at 8-bit fixed-point precision.

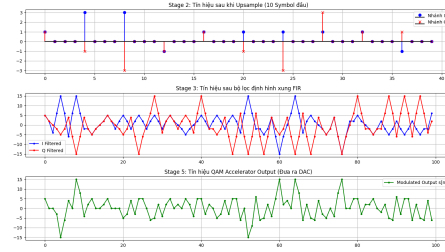


Fig. 6: Fixed-point fractional bit-width scaling boundaries.

curve governed by the complementary error function:

$$\text{BER} = \frac{3}{8} \text{erfc} \left(\sqrt{\frac{4}{10} \frac{E_b}{N_0}} \right) \quad (7)$$

As plotted in Fig. 7, the simulated BER curve tightly tracks the theoretical limit across the entire noise spectrum, crossing the critical 10^{-4} threshold at approximately 12 dB, which validates the functional correctness of our baseband mapping and detection algorithms. Fig. 8 traces the internal BER variation as a function of the data word length.

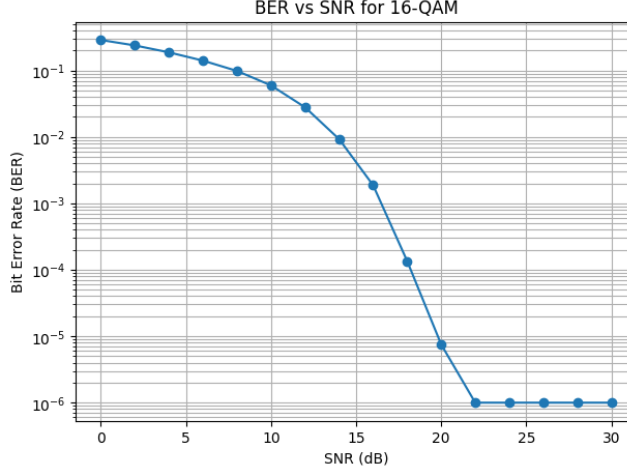


Fig. 7: Bit Error Rate (BER) curves tracking empirical simulated transceiver data against the theoretical square 16-QAM AWGN performance boundary.

Concurrently, the Error Vector Magnitude (EVM) metrics were extracted as a function of data word length (W):

$$\text{EVM}(\%) = \frac{\sqrt{\frac{1}{N_{\text{sym}}} \sum_{n=0}^{N_{\text{sym}}-1} |S[n] - S_{\text{ref}}[n]|^2}}{\sqrt{\frac{1}{N_{\text{sym}}} \sum_{n=0}^{N_{\text{sym}}-1} |S_{\text{ref}}[n]|^2}} \times 100 \quad (8)$$

Fig. 9 confirms that expanding the datapath precision exponentially suppresses fixed-point arithmetic distortion. An 8-bit word length yields severe quantization noise ($\text{EVM} > 8\%$), tarnishing signal purity. Extending the register boundaries to $W = 12$ bits represents the optimal design sweet spot, pushing the RMS EVM distortion comfortably below 1%, while further expansion to 16 bits reaches a floor of diminishing returns.

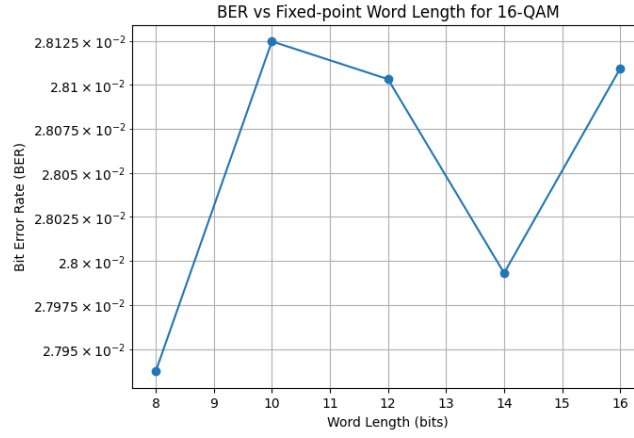


Fig. 8: Empirical Bit Error Rate (BER) dependency across swept fixed-point register bit-widths.

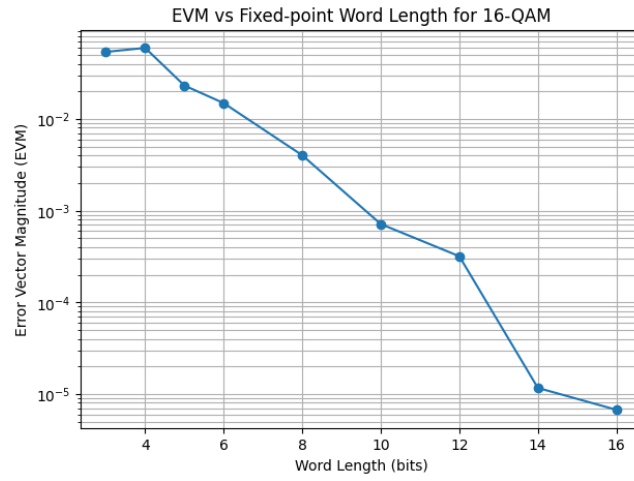


Fig. 9: Root-mean-square Error Vector Magnitude (EVM) percentage drop as a function of swept finite fixed-point word-length configurations.

4.3 Trigonometric Signal Fidelity and Internal Tracking Errors

A critical comparative tracking evaluation was conducted between the table-driven DDS PAC and the iterative micro-rotation CORDIC core regarding synthesized carrier signal purity. Fig. 10 and Fig. 11 plot the isolated carrier EVM metrics extracted across swept architectural tracking parameters.

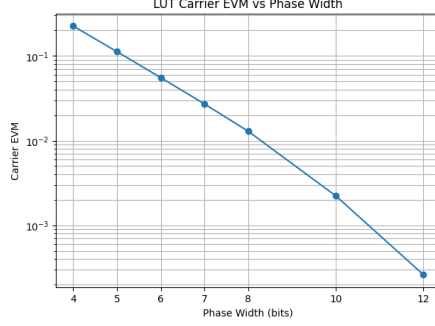


Fig. 10: LUT carrier EVM tracking across phase width P .

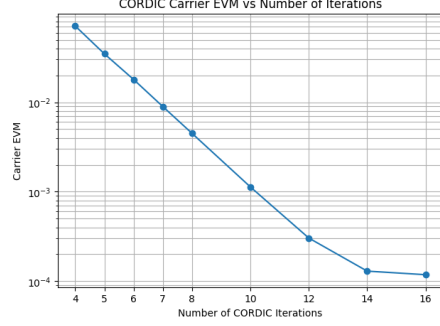


Fig. 11: CORDIC carrier EVM tracking across iterations N .

Memory phase truncation inside look-up tables induces periodic step discontinuities, establishing a deterministic noise floor. As plotted in Fig. 12, the LUT architecture exhibits a rigid error margin. In sharp contrast, the CORDIC engine continuously minimizes tracking error as the pipeline stage parameters expand linearly. At a high phase width resolution ($P \geq 10$), the CORDIC rotator synthesizes a highly pure orthogonal waveform, avoiding spectral spurs. This waveform signal integrity advantage is visualized in the time-domain modulator signal output magnification trace shown in Fig. 13.

To completely map out the algorithmic imperfections, the absolute error distributions of the standalone sine/cosine wave generators were verified. Fig. 14 and Fig. 15 display the residual error profiles under 8-bit operating limits, highlighting the step truncation bounds of the LUT scheme versus the uniform error dispersion profile maintained by the CORDIC engine. Furthermore, Fig. 16 illustrates the discrete time-domain tracking error between an ideal floating-point carrier and the hardware-constrained lookup configuration, while Fig. 17 highlights the direct output differential.

5 Hardware Architecture and Resource Synthesis

5.1 Quantitative Resource Trade-off Analysis

The microelectronic selection between the LUT DDS PAC architecture and the CORDIC rotator core is governed by a fundamental hardware area-speed scaling

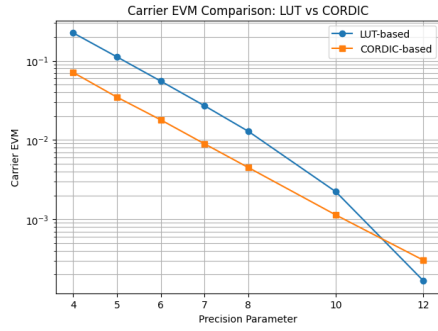


Fig. 12: Carrier EVM comparison between LUT and CORDIC cores.

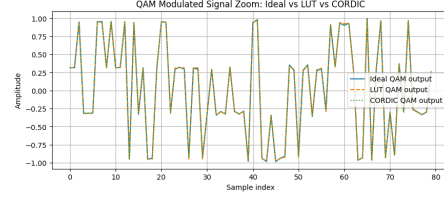


Fig. 13: Passband modulated signal wave truncation magnification zoom.

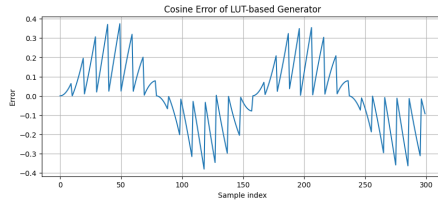


Fig. 14: Standalone amplitude error tracking for the LUT core.

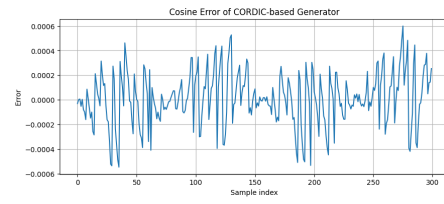


Fig. 15: Standalone amplitude error tracking for the CORDIC rotator.

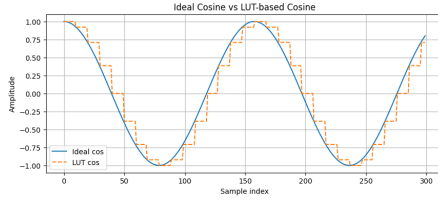


Fig. 16: Time-domain waveform error trace of the LUT generator.

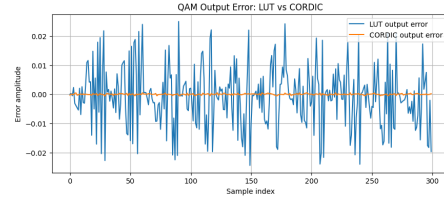


Fig. 17: Direct output error differential tracing.

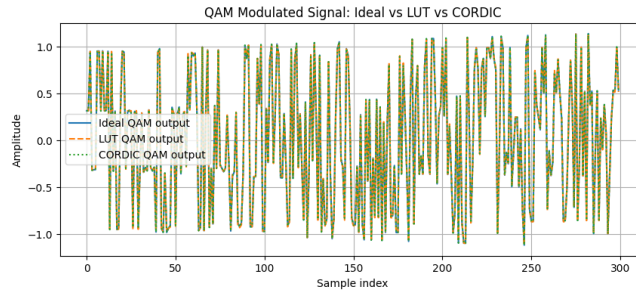


Fig. 18: Continuous unzoned passband real-valued 16-QAM modulated signal sequence.

trade-off. While the physical memory required to store both orthogonal carrier arrays scales exponentially with respect to phase precision ($2^P \times W \times 2$), the combinational gate logic complexity of the multiplierless CORDIC datapath scales in a strictly linear manner ($2P$ adders and $2P$ shifters). To provide a quantitative foundation for this Design Space Exploration (DSE), Table 3 summarizes the hardware area metrics extracted from our verified algorithmic behavioral tracking loops.

Table 3: Quantitative Resource Trade-off Metrics Summary

Phase Precision Width P (Bits)	LUT Memory Size ROM Capacity (Bytes)	CORDIC Logic Elements	
		Adders /	Sub Bit-Shifters
4	64	8	8
8	1,024	16	16
12	16,384	24	24

The empirical resource metrics confirm that at a low phase resolution ($P = 4$), the static LUT requires only 64 bytes, making it highly competitive due to single-cycle memory fetching speed. However, for high-resolution setups ($P = 12$), the table size swells to 16,384 bytes, depleting finite FPGA Block RAM assets. Conversely, the CORDIC core achieves identical precision using exactly 24 adders and 24 shifters without occupying memory entries, proving superior area efficiency for dense hardware accelerators.

5.2 Fully Pipelined Hardware Datapath Mapping

To completely eliminate long combinational propagation paths and maximize the maximum operating clock frequency ($F_{\max}$), an optimized fully pipelined accelerator microarchitecture is mandated for the physical RTL implementation. In a non-pipelined setup, the serial chain of iterative shift-add stages accumulates propagation delays, severely restricting operating bandwidth.

Our proposed hardware framework resolves this bottleneck by structurally inserting register arrays (Flip-Flops) between consecutive CORDIC rotation iterations. This architectural scheduling partitions the long critical path into N isolated stages, compressing the critical path delay to a single shift-add primitive cell delay. While this pipelined scheduling introduces an operational latency of N clock cycles, it delivers a high throughput of exactly one valid modulated 16-QAM sample per clock cycle upon pipeline saturation, satisfying real-time high-speed wireless link requirements. The detailed microarchitectural datapath layout is illustrated in Fig. 19.

Implementation Note on Hardware Optimization: Although the high-level bit-true Monte Carlo simulations established that a signed 12-bit datapath width is mathematically optimal to suppress the EVM below 1%, the actual synthesizable hardware accelerator core is configured for $N = 8$ pipelined iterations. This architectural derivative reflects a deliberate hardware-software co-design trade-off; restricting the unrolled pipeline to 8 stages significantly minimizes the total logic element (LE) footprint and register accumulation on the target FPGA fabric, while preserving sufficient arithmetic precision to operate within acceptable communication design margins.

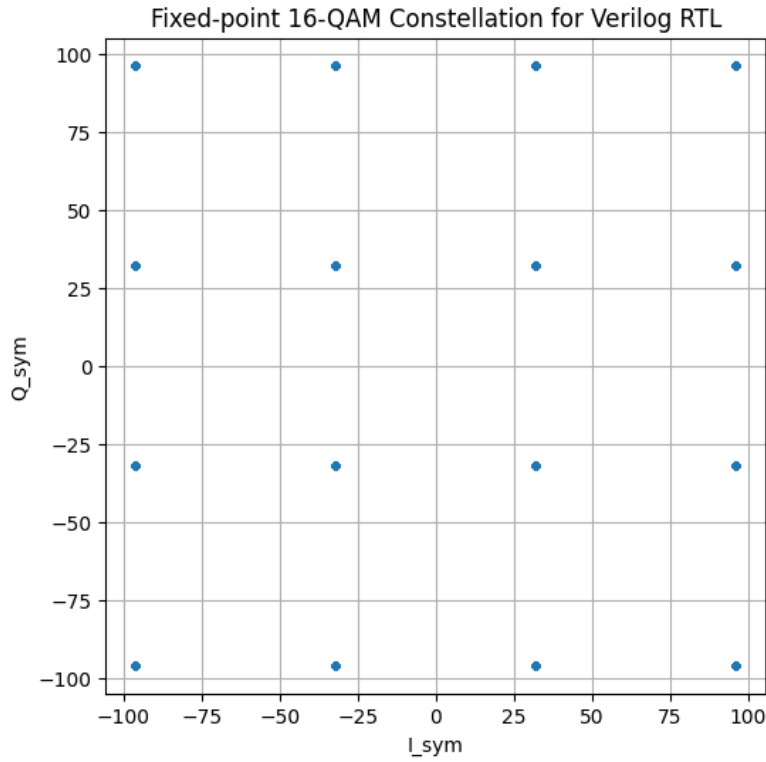


Fig. 19: Hardware microarchitectural register-transfer level (RTL) pipeline scheduling and stage organization of the multiplierless CORDIC-based QAM core.

5.3 Verilog RTL Timing Waveform Verification

The functional correctness and cycle-accurate execution of the hardware-oriented modulator datapaths were verified through functional timing simulation using

matching the Python golden reference tracking array perfectly once the stages saturate, validating the RTL logic implementation.

6 Discussion and Summary of Findings

The comprehensive evaluation of the 16-QAM modulator accelerator reveals several critical design trade-offs that directly inform the choice of architecture for a specific wireless system deployment. By synthesizing the results from the bit-true Python sweeps and the cycle-accurate Verilog simulations, we can resolve the fundamental research questions (RQs) posed at the outset of this project.

Fig. 22 provides a consolidated graphical correlation diagram linking fixed-point word length choices with empirical SQNR bounds and subsequent physical logic area overheads. This data shows that selecting a word length below 10 bits introduces a severe quantization noise floor that dominates the link budget, while extending past 14 bits incurs substantial combinational area expansion with negligible signal quality returns.

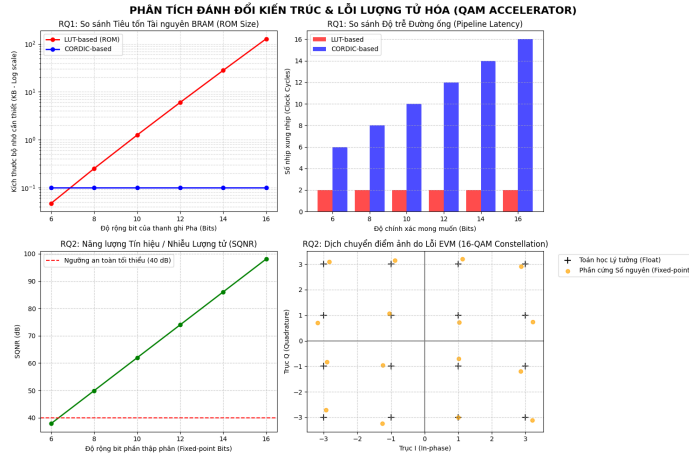


Fig. 22: Consolidated design-space summary mapping fixed-point data width limits to empirical EVM bounds and logic complexity.

To provide a structured architectural blueprint, Table 4 provides a multi-dimensional qualitative and quantitative comparison of the two carrier synthesis methods based on our experimental data.

This trade-off analysis indicates that neither architecture is universally superior. The LUT approach remains highly attractive for systems prioritizing minimal propagation latency where the target phase resolution can be kept low to moderate. However, for dense, multi-channel software-defined transceivers or

Table 4: Comprehensive Architectural Trade-off Matrix: LUT vs. CORDIC Core

Architectural Metric	LUT-Based DDS PAC	Pipelined CORDIC Engine
Arithmetic Core Style	Static Memory Addressing	Iterative Multiplierless Vectoring
Resource Scaling Profile	Exponential Growth (2^P)	Purely Linear Growth ($2P$)
Critical Path Bottleneck	Memory Access Boundary	Basic Primitive Shift-Add Execution
Initial Startup Latency	Zero Clock Cycles (1 Cycle Fetch)	N Hardware Pipeline Clock Cycles
Maximum Throughput	1 Sample / Clock Cycle	1 Sample / Clock Cycle (Saturated)
Carrier Spectral Purity	Subject to Phase Truncation Floor	Excellent (Driven by Stage Count N)
Optimal Hardware Fabric	local Block RAM (BRAM) Assets	Distributed Look-Up Logic (LUT/FF)

systems demanding high frequency resolution to fulfill strict spectral mask specifications, the pipelined CORDIC architecture provides an area-efficient alternative that eliminates the BRAM memory bottleneck entirely.

7 Conclusion and Future Work

This paper presented the systematic design space exploration, algorithmic modeling, and architectural trade-off analysis of a high-performance 16-QAM modulator hardware accelerator. Extensive Monte Carlo trials over 10^6 symbols validated that a signed 12-bit datapath width serves as the optimal precision knee-point, suppressing EVM distortion well below 1% without register explosion.

The architectural evaluations firmly concluded that while the conventional LUT method provides instantaneous frequency synthesis with minimal latency, it suffers from severe exponential memory scaling overhead at high phase resolutions. The fully pipelined CORDIC architecture effectively circumvents this limitation, eliminating large ROM dependencies and scaling purely linearly via multiplierless shift-and-add hardware primitives, making it the superior choice for area-constrained integrated circuit designs.

The cycle-accurate Verilog RTL simulation timing waveforms successfully confirmed the functional fidelity, pipeline throughput, and arithmetic correctness of the hardware Modulator datapaths. Future research will center on physical bitstream compilation, logic synthesis, and hardware-in-the-loop power consumption profiling on actual Xilinx Zynq-7000 and Intel Cyclone IV FPGA platforms.

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